Mathematical Modeling and Signal Processing of UAV Bearing Faults

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SC24B100

October 10, 2026

Bearing Fault Kinematics

Unlike mass unbalance, bearing defects generate high-frequency mechanical impacts dictated by the internal geometry of the bearing. Assuming the outer ring is fixed to the motor casing, the inner ring rotates with the shaft without slip, and d/D is a dimensionless diameter ratio, the specific characteristic frequencies for different types of bearing damage are defined as follows:

Bearing inner ring damage:

$$f_i = \frac{Z}{2} f_r \left(1 + \frac{d}{D} \cos \alpha\right) \tag{1}$$

Bearing outer ring damage:

$$f_o = \frac{Z}{2} f_r \left(1 - \frac{d}{D} \cos \alpha\right) \tag{2}$$

Bearing Rolling element damage:

$$f_b = \frac{D}{2d} f_r \left(1 - \frac{d^2}{D^2} \cos^2 \alpha\right) \tag{3}$$

in which:

- Z: Number of rolling elements.
- d: Rolling element diameter.
- D: Pitch diameter (diameter of the circle describing the center of the rolling elements).
- α: Contact angle in radians.
- f_r: Rotation frequency of the shaft in Hertz.

As a kinematic unit test for the simulation architecture, the relationship $f_i + f_o = Z f_r$ must hold true. Furthermore, because a damaged rolling element strikes both the inner and outer races during each of its own revolutions, its physical impact rate is $2f_b$.

Physical Parameterization of Impact Impulse

A bearing defect generates periodic micro-collisions. When a rolling element of mass m_b strikes a defect of width w_d, the collision physics can be modeled using a linearized contact stiffness K_h.

The impact velocity v_{impact} of the rolling element into the trailing edge of the spall is a lumped assumption:

$$v_{impact} = c_k \cdot \Omega \cdot w_d \tag{4}$$

Where Ω is the angular velocity of the rotor ($2\pi f_r$) and c_k is a kinematic constant absorbing the bearing geometry.

Modeling the collision as a mass m_b striking a linear spring K_h, the ball compresses the spring and rebounds. The resulting force follows a half-sine wave shape for exactly half a period, $F(t) = F_{max} \sin(\pi t/\Delta t)$, with a peak force of $F_{max} = v_{impact}\sqrt{m_b K_h}$ and a contact duration of $\Delta t = \pi\sqrt{m_b/K_h}$.

Given typical micro-bearing parameters, Δt is on the order of microseconds. Because this duration is far shorter than the sampling period of a standard flight controller Inertial Measurement Unit (IMU), the collision is mathematically modeled as an instantaneous Dirac impulse. The area under the half-sine force curve dictates the exact momentum change (impulse I) delivered to the structure:

$$I = \int_0^{\Delta t} F_{max} \sin\left(\frac{\pi t}{\Delta t}\right) dt = \frac{2}{\pi} F_{max} \Delta t = 2 m_b v_{impact} \tag{5}$$

This establishes that the impulsive weight injected into the simulation scales strictly with the change in momentum of the rolling element, bypassing the need to parameterize unobservable microsecond durations.

Directional Vectorization and Defect Modulation

Standard UAV brushless motors utilize deep-groove ball bearings with a nominal contact angle of $\alpha \approx 0$. Consequently, the axial (Z-axis) component of the impact force is negligible ($F_z \approx 0$), and the impact energy is released entirely into the radial X and Y axes.

The spatial projection of this radial force depends strictly on the physical location of the spall:

- **Outer Race Defect:** The defect is stationary on the motor casing at a constant angle β. The impacts have uniform amplitude.
- **Inner Race Defect:** The defect rotates with the shaft. Its angular position is time-dependent, defined as $\beta(t) = \Omega t + \beta_0$. Furthermore, the defect passes in and out of the primary load zone once per revolution, causing the impact amplitude to be modulated at the shaft running speed f_r.

To ensure realistic spectral generation, a perfectly periodic impulse train is insufficient. A random slip jitter ϵ_n (approximately 1-2% of the fault interval $T_{fault} = 1/f_{fault}$) must be added to account for rolling element slide.

The resulting localized time-domain forcing functions for the structural radial axes are:

For an Outer Race Defect (Stationary β):

$$F_{o,x}(t) = I\cos(\beta)\sum_{n=0}^{\infty}\delta(t - nT_o - \epsilon_n) \tag{6}$$

$$F_{o,y}(t) = I\sin(\beta)\sum_{n=0}^{\infty}\delta(t - nT_o - \epsilon_n) \tag{7}$$

For an Inner Race Defect (Rotating β, Amplitude Modulated):

$$F_{i,x}(t) = [I\cos(\Omega t + \beta_0)]\sum_{n=0}^{\infty}\delta(t - nT_i - \epsilon_n) \tag{8}$$

$$F_{i,y}(t) = [I\sin(\Omega t + \beta_0)]\sum_{n=0}^{\infty}\delta(t - nT_i - \epsilon_n) \tag{9}$$

In the resulting envelope spectrum, this amplitude modulation will produce clear sidebands at $f_i \pm f_r$ for inner race faults, and ball defects will exhibit sidebands at the cage frequency.

Structural LTI Convolution and IMU Realities

The continuous forcing functions are passed through the Linear Time-Invariant (LTI) structural transfer function $H(\omega)$ of the UAV chassis. The output acceleration $a(t)$ is the convolution of the input force and the structure's impulse response $h(t)$, superimposed over the baseline rigid-body acceleration and parameterized Gaussian noise $\eta(t)$:

$$a_x(t) = a_{rigid_body,x}(t) + \left(\int_{-\infty}^{\infty} h_x(\tau)F_{bearing,x}(t-\tau)d\tau\right) + \eta_x(t) \tag{10}$$

This continuous output must then be discretely sampled to match the specifications of the flight controller IMU, requiring a predefined sample rate (f_s) and Nyquist bandwidth limit ($f_s/2$) to accurately reflect the hardware constraints of the fault detection algorithm.

Signal Processing Pipeline and Envelope Analysis

Because bearing impacts are low-energy events buried within broadband mechanical noise, the structural ringing must be demodulated using Envelope Analysis. The signal processing pipeline executes the following sequence:

1. **Band-Pass Filtering:** A band-pass filter is applied to the raw discrete acceleration signal. The passband is explicitly centered around the known structural resonance peak of the chassis transfer function $H(\omega)$, isolating the high-frequency structural carrier wave while discarding low-frequency mechanical unbalance and rigid-body dynamics.

2. **Analytic Signal Construction:** The Hilbert Transform $\mathcal{H}\{a_{filtered}(t)\}$ is applied to shift the phase by 90°, constructing the complex analytic signal $z(t) = a_{filtered}(t) + j \cdot \mathcal{H}\{a_{filtered}(t)\}$.

3. **Envelope Extraction:** The instantaneous magnitude of the analytic signal is calculated to extract the time-domain impact envelope $E(t) = |z(t)|$.

4. **Mean Subtraction:** To prevent the DC component from dominating the final spectrum, the mean of the envelope is subtracted: $E_{AC}(t) = E(t) - \mu_E$.

5. **Fourier Transform:** The Fast Fourier Transform (FFT) is applied to $E_{AC}(t)$ to yield the final Envelope Spectrum, isolating the precise kinematic defect frequencies and their respective modulation sidebands.